

Biodiversity

Species: roles; Types: extinct, endemic, endangered and rare species

Roles of species: Each species plays a specific ecological role called its ecological niche.

The roles of the species are classified into 5 types.

1. Native species
2. Non-native species
3. Indicator species
4. Keystone species
5. Foundation species

1. Native species: Those that live and develop in a particular ecosystem.

E.g.: Lions in the forest,

2. Non-native species: Those that are either accidentally or forcefully introduced into a different environment. These may be migratory as well.

E.g.: Domesticated species such as cattle, chicken.

3. Indicator species: Those that provide information about the change in the environment and climate of a particular ecosystem.

Amphibians, like frogs, toads and salamanders, are known as indicator species. They are extremely sensitive to changes in the environment and can give scientists valuable insight into how an ecosystem is functioning.

4. Keystone species: Those that are abundant and have a great effect on other species. When the activities of a species determine community structure that species is called keystone species. For example, consider the case of the starfish, *Pisaster ochraceus*. When this starfish removed from the rocky intertidal areas of western north America, the mussel *Mytilus californianus* was able to occupy the space and excluded other invertebrates and algae which require attachment sites. However, under natural conditions, predation of mussels by starfish keeps their population under control and does not allow it to become dominant. This permits other species requiring attachment sites to survive in such habitats. Other examples of a keystone species could be of the African elephant wolves, leopards, alligators.

Keystone species may be relatively rare in natural communities or may not be easily recognised. At present, few terrestrial communities are believed to be organised by keystone species, but in aquatic community's keystone species may be common.

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5. Foundation species: Those that have a large contribution towards creating and maintaining habitats that support other species.

E.g.: corals, earthworms,



Fig: - Corals

Types of species:

1. Endemic
2. Extinct
3. Endangered
4. Rare
5. Exotic

1. Endemic species:

- Native to a particular place. E.g.: Asiatic Lion, Red Panda
- Endemic species are those plants/animals unique to a defined geographic location, such as an island, nation, or a defined zone or habitat type.
- Areas containing endemic species are often isolated in some way preventing easy spread of species to other areas
- E.g., Islands in Hawaii, New Zealand and southern tip of Africa contain almost 90% endemic species
- Due to the geographic restrictions of such species, endemic species are often endangered

2. Extinct species:

- Species where the last remaining member of the species has died, or is presumed beyond reasonable doubt to have died.
- If not seen for 50 years. Used for species which are no longer known to exist in the world. E.g.: Dinosaur, Dodo, Himalayan Quail, Indian Cheetah Extinction of an animal or plant occurs when no more individuals of that species are alive anywhere in the world.
- This is a natural part of evolution, but certain extinctions happen at a much faster rate.

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- E.g., the end of the Cretaceous period 65 million years ago saw a mass extinction that caused the death of several plants and animals, including dinosaurs.
- Why this extinction happens so fast. Human interference in the form of hunting, over-exploitation and habitat destruction is also causing rapid extinction.

3. Endangered/vulnerable species:

- Endangered or threatened species is one that is considered at the risk of extinction.
- A species can be listed as endangered at the state, national or international level.
- We can save those species if we identify them in the early stage.

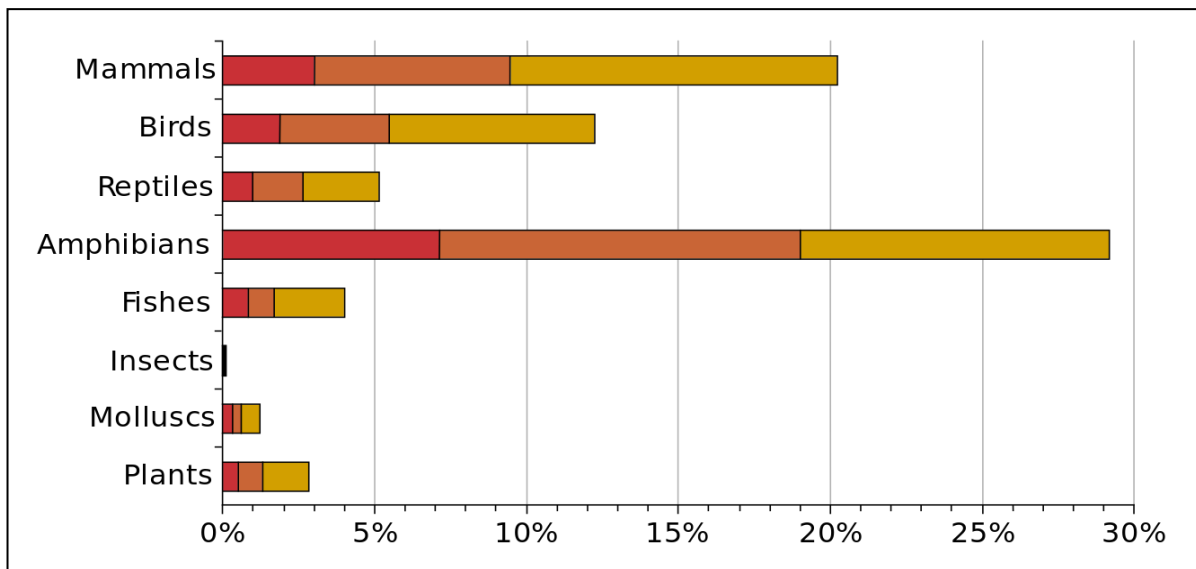


Fig: - Endangered species

4. Rare species:

- Species with small world populations that are not at present endangered or vulnerable but are at risk. E.g.: Sparrows, black buck.
- This is distinct from the term endangered or threatened. Rare species are a group of organisms that are uncommon, scarce or infrequently encountered.
- They are normally species with small populations, and several move into the endangered category if negative factors operate against them

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5. Exotic species:

- Non- native species which have been moved by human being from their native place to non-native environment. E.g.: Orchids, Cacti.
- Exotic species are a group of organisms that are non-native, moved into the particular area by humans from their native environment.
- E.g., orchids, cacti, caged animals in zoos, etc.



Species interactions:

Biological interaction is the effect that a pair of organisms living together in a community have on each other. They can be of the same species (intraspecific) or different species (interspecific).

The effects may be **short-term** like pollination and predation or **long-term**, strongly influencing the evolution of the other species.

Important types of interactions:

- Predation
- Parasitism
- Competition

Predation:

- In addition to competing for food or space, species in a community may interact by predation which literally means plundering.
- Requires one organism, the predator, to kill and eat another organism, its prey.
- In most cases, both are animals, both of different species (inter-specific); but if both are of same species, (intra-specific), it is called cannibalism.
- It is a short-lived interaction, but very durable in terms of influence on the evolution of both partners, resulting in both partners co-evolving.
- predators have sharp claws or jaws to grip the prey, with other adaptations to improve hunting efficiency.

Ex: Crocodiles are some of the evolutionarily oldest and dangerous predators

The effect of predation on population has been studied theoretically and practically because it has economic implication for our own species. Predation may affect populations mainly in three ways: restricts distribution or reduces abundance affects structure of community is a major selective force, and many adaptations that we see in organisms such as mimicry or warning colouration have their explanation in predator - prey coevolutions.

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2. Parasitism:

Parasitism is a relationship between species where one organism, the parasite, lives on/in another organism, the host, causing it harm, and is adapted structurally to this way of life.

Ex: A tick living on a dog is an example of parasitism. In this relationship, the tick gains a food source by drinking the dog's blood.

3. Competition:

- Competition is the interaction between organisms where the fitness of one is lowered by the presence of the other.
- Competition occurs over resources. For plants light, nutrients, and water may be important resources. Plants may compete for pollinators or for attachment sites. Water, food and mates are possible resources for animals, and they may compete for space such as nesting sites, wintering sites or places that are safe from predators. Thus, we see that resources can be complex and diverse.
- Competition is often for a resource such as water, food, territory or access to females for reproduction.
- There are two types of competitive interactions: Exploitative or scramble competition occurs when a number of organisms of same or different species utilise common resources that are in short supply.
- Interference or contest competition occurs when organisms seeking a resource will harm one another in the process even if the resource is not in short supply.
- Competition could be intra-species competition or inter-species competition.



Fig: - Inter-species competition



Fig: - Intra-species competition

- According to evolution, the species less suited to compete for resources either adapts or dies out; competition plays an important role in natural selection.